

William “Billy” Bratz

Senior software engineer with 11 years building distributed systems and, since late 2025, production LLM agent systems. I founded the MCP server and orchestrator that became my organization's LLM platform, and I turn difficult operational lessons into durable systems, standards, and writing.

Email LinkedIn GitHub williambratz.com

EXPERIENCE

Carvana

2020–Present · Remote

Senior Software Engineer II

LLM Platform: Conversational AI

Jun 2026–Present · Python, FastAPI, FastMCP, Kubernetes, Datadog

- Operate an MCP platform serving more than 10 million weekly tool calls, connecting conversational AI to production systems.
- Root-caused a multi-service outage in minutes across a five-hop request chain, halting a harmful redeploy and directing the recovery.
- Demonstrated that an incident remained active after apparent recovery because of timeout asymmetry, changing how the team validates service health.
- Designed and shipped a same-day in-cluster route that delivered a double-digit p95 latency reduction while preserving the public path as a safe fallback.
- Found and corrected a CI isolation defect across 383 tests, restoring zero-network unit-test guarantees.
- Resolved an MCP HTTP 421 outage using APM evidence, then added a dependency-host guard, pinned lockfile, monitors, and an audit of sibling services.
- Eliminated order-of-magnitude retry amplification and validated the change under live production traffic.
- Shipped three customer-context MCP tools, including the platform's first MCP-to-MCP workflow, within service-level objectives.
- Used production log analysis to isolate and correct a timestamp race, then verified the fix under load.
- Designed a split test for a model migration covering a significant share of assistant traffic.

Verifications & Underwriting

2020–Jun 2026 · .NET, Azure, Kafka, Kubernetes

- Founded an MCP server and orchestrator in December 2025. I wrote roughly 6,650 lines in six days and supported it with a 445-line operator guide. It became the foundation of the organization's LLM platform.
- Built an autonomous delivery pipeline that completed 17 runs over 10.6 hours and produced 20 reviewable commits.
- Created an internal marketplace of nine agent plugins and introduced recursive language-model techniques for work beyond a model's normal context window.
- Designed the operating contract for a custom Claude Slack integration used in production investigation and engineering workflows.
- Architected a fraud-analysis service with 13 ADRs, C4 diagrams, and five runbooks to support a sixfold growth target.
- Led a workload-identity migration spanning five shared packages and more than five services; wrote the 1,000-line implementation playbook used by other teams.
- Reduced advocate insurance work from more than 20 minutes to under four, and delivered an OCR proof of concept in three days with an error rate below 2%.
- Cut one service's memory ceiling from over 1.5 GB to under 243 MB, reducing total memory utilization from 60–80% to below 10%.
- Contributed across services to a top company rearchitecture supporting co-buyers, while mentoring engineers on domain workflows and dependencies.
- Authored 15 architecture decisions and multiple engineering standards and root-cause analyses. Opened 143 pull requests and reviewed 177 across 20 repositories in 28 weeks; 58% of the work was self-scoped.

CoStar Group (formerly STR)

2015–2020

Software Engineer

- Served as technical lead for Instant Insights, decomposing monolithic services into reusable .NET microservices and building a React and Redux analytics experience.
- Joined a three-person team integrating STR's global hospitality data into CoStar Suite, a product serving more than one million daily searches.
- Built a report-generation platform with AWS Lambda and Step Functions, plus reusable translation and currency-conversion modules.

WRITING & OPEN SOURCE

WilliamBratz.com

Nine essays on production engineering and AI systems, including "Production Reveals the System," "From Chatbots to Persistent Organizational Memory," and "Engineering in the Age of Agents."

vaultkit

A personal, LLM-maintained knowledge engine: roughly 6,400 lines of Python, more than 320 tests, three scheduled maintenance cadences, and a scaffold adopted by four teams. The reusable vault-scaffold workflow is available on GitHub.

Independent products and open source

Built Neverending Story Pointer, a free real-time estimation app, and How Unique Is a GUID?, an interactive probability experience also available for UUIDs. Contributions to RestSharp and NUnit.

TECHNICAL SKILLS

Languages

Python, C#, TypeScript, JavaScript, SQL, GraphQL

AI & agents

Agent orchestration, MCP server/client/gateway federation, FastMCP, OpenAI and Anthropic APIs, PydanticAI, tool calling, structured output, RAG with Pinecone, prompt experiments, evaluations, observability, Claude plugins and skills

Backend

FastAPI, asyncio, Pydantic v2, .NET Core, REST, JSON-RPC, gRPC, Kafka, Azure Service Bus, event-driven systems

Data

Cosmos DB, Redis, PostgreSQL, Azure SQL, SQL Server, DynamoDB

Platform

Kubernetes (AKS/GKE), Docker, Helm, Tekton, Flux, GitHub Actions, Azure DevOps, Lambda, Step Functions, Workload Identity, OIDC

Observability

Datadog, Splunk, Prometheus, Grafana, OpenTelemetry, SLOs, Locust

EDUCATION

Arizona State University

B.S. Software Engineering · In progress

Volunteer State Community College

A.A.S. Computer Programming · 2022

Phi Theta Kappa · Dean's List